Adarsh Singh

Software Engineer | AI & Security

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Education

Lovely Professional University, Jalandhar, Punjab — **August 2022 – July 2026**
Bachelor of Technology — Computer Science and Engineering (Minor: Cyber Security)
CGPA: 7.27

Experience

Titan Company Limited — *Jan 2026 – Aug 2026*
Intern | InfoSec Department — *Onsite, Bengaluru*

WAF Automation · Node.js, Playwright, JavaScript · *AI-assisted*

- Cut the first run (report download + SwyftComply request) across **170+ WAF-protected domains** from **3–4 hours to 45 min–1 hour** with headless Node.js/Playwright. A run 72 hours later re-downloads reports to confirm fixes.
- Added SSO-aware Edge profiles, dropdown discovery, session keep-alive, skip-if-already-requested logic for safe re-runs, and a monthly CSV run log. Team runs it twice a month.

VAPT Assessment Portal · React, TypeScript, Node.js, Express, Prisma, MySQL · *AI-assisted*

- Cut report creation from **3–4 hours to 30–60 minutes** with a portal replacing manual Word and Excel workflows for web, mobile and API assessments, used by the VAPT team.
- Integrated the company's LLM gateway (JWT auto-refresh) to suggest CVSS/CWSS scores, CWE IDs and remediation text from analyst PoCs. Analysts approve every AI field; PoC evidence is kept for revalidation.
- Added append-only audit tables, SLA breach tracking, risk heatmaps, and async PDF via BullMQ and Puppeteer.

Large-Scale Data Diffing Tool · React, TypeScript, Node.js, TanStack Virtual, Zustand, Microsoft Graph API · *AI-assisted*

- Built a dashboard ingesting two Excel reports (**up to 100MB each**) and classifying **396,000+ AWS Inspector findings** as new, resolved, or common, replacing the manual Excel comparison. Rows render via TanStack Virtual.
- Fixed a parsing bug where scientific notation made two AWS account IDs look identical and hid one account's findings.
- Integrated Microsoft Graph API to email per-account Excel reports to owners, with a session log of reports sent.

Projects

Adarshclaw — Secure Human-in-the-loop AI Assistant — *Oct 2026*
TypeScript, Node.js, Bun, Vercel AI SDK, Docker

- Engineered a secure AI coding assistant supporting autonomous task execution, codebase analysis, and isolated web research featuring a **human-in-the-loop approval workflow** and **Telegram remote control**.
- Architected defense-in-depth security, routing AI shell execution into **network-disabled Docker containers** (`oven/bun:1`) with dropped Linux capabilities and **post-execution** `git diff` **snapshot auditing**.
- Integrated a **secondary AI verification agent** to automatically audit staged mutations for prompt injection defense, enforcing strict tool isolation policies and **10-minute task runtime locks**.

Snip.ly — URL Shortener — *Oct 2025*
React, Vite, Tailwind CSS, Cloudflare Workers, D1 (SQLite)

- Built a serverless URL redirection service on Cloudflare Workers (V8 isolates) and D1 (edge SQLite), delivering **sub-50ms redirects at the edge**.
- Hardened the API against SSRF with destination-host validation (blocking localhost, loopback, and RFC 1918 private IP ranges), origin-allowlisted CORS with `Vary: Origin`, and JWT verification via `jose` (Web Crypto).

Certifications

Oracle Cloud Infrastructure 2025 Generative AI Professional — Oracle — *Oct 2025*
Oracle Cloud Infrastructure 2025 Certified Developer Professional — Oracle — *Sep 2025*
Claude 101 — Anthropic — *Mar 2026*
Claude Code in Action — Anthropic — *Apr 2026*

Technical Skills

Languages: C/C++, TypeScript, JavaScript, Java, Python, SQL
Frameworks & Libraries: React, Node.js, Express.js, Bun, FastAPI, Tailwind CSS, TanStack Virtual, Zustand, Vercel AI SDK
Databases & Messaging: MySQL, PostgreSQL, SQLite (D1), MongoDB, Prisma ORM, BullMQ
Cloud, Security & DevOps: Docker, Linux, Cloudflare Workers, Playwright, Puppeteer, Git, GitHub